

Fibromyalgia: an overview.

PMID: 19962494 | Indexed in PubMed

Fibromyalgia is the diagnosis given to individuals with chronic widespread musculoskeletal pain for which no alternative cause, such as tissue inflammation or damage, can be identified. Fibromyalgia is now believed to be, at least in part, a disorder of central pain processing that produces heightened responses to painful stimuli (hyperalgesia) and painful responses to nonpainful stimuli (allodynia). Aberrations in central pain processing may also be partly responsible for symptoms experienced in several chronic pain disorders that coaggregate with fibromyalgia, which is itself a product of genetic and environmental factors. Thus, aberrational central pain processing is implicated in irritable bowel syndrome, temporomandibular disorder, chronic low back pain, and certain other chronic pain disorders. Fibromyalgia and related disorders appear to reflect deficiencies in serotonergic and noradrenergic, but not opioidergic, transmission in the central nervous system. The heightened state of pain transmission may also be owing to increases in pronociceptive neurotransmitters such as glutamate and substance P. In some cases, psychological and behavioral factors are also in play. Although the overlapping symptomatology between fibromyalgia and related disorders may present diagnostic challenges, proper examination and observation can help clinicians make an accurate diagnosis. In recent years, the vastly improved understanding of the mechanism underlying fibromyalgia and the related spectrum of diseases has fostered rapid advances in the therapy of these chronic pain disorders by both pharmacologic and nonpharmacologic interventions.

Fibromyalgia: presentation and management with a focus on pharmacological treatment.

PMID: 19225604 | Indexed in PubMed

Fibromyalgia is a condition with widespread muscle pain. Prevalence studies showed that 2% to 7% of the population have fibromyalgia, which affects approximately one million Canadians. Fibromyalgia is most common in women, but it also involves men and children. As with most chronic illnesses, the causes of fibromyalgia are unknown. However, recent research supports underlying abnormalities in the central nervous system, which supports fibromyalgia as a chronic disease state and valid clinical entity. Pain is the primary symptom, often accompanied by overwhelming fatigue, sleep dysfunction and cognitive impairment. In 1990, the American College of Rheumatology developed diagnostic criteria for the diagnosis of fibromyalgia. Lifestyle changes, including pacing of activities and aerobic exercise, are very important in managing fibromyalgia symptoms. Emotional and behavioural therapy can also be helpful. Controlled trials of antidepressants, gabapentinoids, tramadol, zopiclone and sodium oxybate have shown effectiveness in fibromyalgia patients. Pregabalin and duloxetine were recently approved in the United States. Effective management of fibromyalgia is complex and requires a multidisciplinary treatment approach. Response and tolerance of different therapeutic interventions vary from patient to patient. Recent advances in the pathophysiology of fibromyalgia offer hope for new and improved therapies in the management of this disabling condition.

Should rheumatologists be managing fibromyalgia?

PMID: 22408215 | Indexed in PubMed

Fibromyalgia has always struggled to be taken seriously. The vague mixture of aches, pains, stiffness and fatigue with no clear clinical or investigational findings has led many to doubt its very existence. Evidence is accumulating, however, of demonstrable abnormalities of pain processing and psychosocial factors in fibromyalgia subjects. These may 'prove' its existence, but do they suggest it is rheumatological?

Chronic muscular pain: aetiology and pathogenesis.

PMID: 7850876 | Indexed in PubMed

The conclusion is that no one single mechanism can explain FMS and is thus in that sense a compromise. FMS in some patients may start in the muscle, in other patients in the brain. The combination of peripheral and central factors is the key to the pathogenesis of FMS as long as FMS is defined as a pain syndrome.

Electroacupuncture Reduced Fibromyalgia-Pain-like Behavior through Inactivating Transient Receptor Potential V1 and Interleukin-17 in Intermittent Cold Stress Mice Model.

PMID: 39335365 | Indexed in PubMed

Fibromyalgia (FM) is a widespread musculoskeletal pain associated with psychological disturbances, the etiopathogenesis of which is still not clear. One hypothesis implicates inflammatory cytokines in increasing central and peripheral sensitization along with neuroinflammation, leading to an elevation in pro-inflammatory cytokines, e.g., interleukin-17A (IL-17A), enhanced in FM patients and animal models. The intermittent cold stress (ICS)-induced FM-like model in C57BL/6 mice has been developed since 2008 and proved to have features which mimic the clinical pattern in FM patients such as mechanical allodynia, hyperalgesia, and female predominance of pain. Electroacupuncture (EA) is an effective treatment for relieving pain in FM patients, but its mechanism is not totally clear. It was reported as attenuating pain-like behaviors in the ICS mice model through the transient receptor potential vanilloid 1 (TRPV1) pathway. Limited information indicates that TRPV1-positive neurons trigger IL-17A-mediated inflammation. Therefore, we hypothesized that the IL-17A would be inactivated by EA and TRPV1 deletion in the ICS-induced FM-like model in mice. We distributed mice into a control (CON) group, ICS-induced FM model (FM) group, FM model with EA treatment (EA) group, FM model with sham EA treatment (Sham) group, and TRPV1 gene deletion (*Trpv1*^{-/-}) group. In the result, ICS-induced mechanical and thermal hyperalgesia increased pro-inflammatory cytokines including IL-6, IL-17, TNF α , and IFN γ in the plasma, as well as TRPV1, IL-17RA, pPI3K, pAkt, pERK, pp38, pJNK, and NF- κ B in the somatosensory cortex (SSC) and cerebellum (CB) lobes V, VI, and VII. Moreover, EA and *Trpv1*^{-/-} but not sham EA countered these effects significantly. The molecular mechanism may involve the pro-inflammatory cytokines, including IL-6, IL-17, TNF α , and IFN γ . IL-17A-IL-17RA play a crucial role in peripheral and central sensitization as well as neuroinflammation and cannot be activated without TRPV1 in the ICS mice model. EA alleviated FM-pain-like behaviors, possibly by abolishing the TRPV1- and IL-17A-related pathways. It suggests that EA is an effective and potential therapeutic strategy in FM.

Effectiveness of cognitive behavioral therapy in the treatment of fibromyalgia syndrome: a meta-analytic literature review.

PMID: 22842298 | Indexed in PubMed

Fibromyalgia (FM) is a chronic disorder caused by a dysfunction of central nervous system sensitization. This syndrome is characterized by widespread pain and diffuse tenderness, but often also presents fatigue, sleep disturbances, and a whole range of symptoms such as morning stiffness, decreased physical function and dyscognition. FM is usually treated with pharmacological and non-pharmacological treatments. The non-pharmacological interventions include cognitive behavioral therapy (CBT), physiotherapy, acupuncture and patient education programs. In order to evaluate the efficacy of CBT and compare it with other non-pharmacological treatments, we performed a review of the meta-analytic literature. We evaluated the methodological quality of publications found by following the recommendations of the PRISMA (Preferred Reporting Items for Systematic Reviews and Meta-Analyses) statement. Data showed that CBT does not provide better results than other non-pharmacological treatments on outcomes of pain, fatigue, sleep disturbance and quality of life, at either a short or long-term evaluation. On the contrary, CBT seems to be more effective on symptoms of depression for a short period, whereas it considerably improves the pain self-management and reduces the number of visits to the doctor. The data currently available indicate that cost-effectiveness studies could help us to understand whether the reduction in the number of visits to the doctor could balance the cost of CBT to the health public system.

Associations Between Pain, Current Tobacco Smoking, Depression, and Fibromyalgia Status Among Treatment-Seeking Chronic Pain Patients.

PMID: 25801019 | Indexed in PubMed

As smoking impacts physiological pathways in the central nervous system, it is important to consider the association between smoking and fibromyalgia, a pain condition caused predominantly by central nervous system dysfunction. The objectives were to assess the prevalence of current smoking among treatment-seeking chronic pain patients with (FM+) and without (FM-) a fibromyalgia-like phenotype; test the individual and combined influence of smoking and fibromyalgia on pain severity and interference; and examine depression as a mediator of these processes. Questionnaire data from 1566 patients evaluated for a range of conditions at an outpatient pain clinic were used. The 2011 Survey Criteria for Fibromyalgia were used to assess the presence of symptoms associated with fibromyalgia. Current smoking was reported by 38.7% of FM+ patients compared to 24.7% of FM- patients. FM+ smokers reported higher pain and greater interference compared to FM+ nonsmokers, FM- smokers, and FM-nonsmokers. There was no interaction between smoking and fibromyalgia. Significant indirect effects of fibromyalgia and smoking via greater depression were observed for pain severity and interference. Current smoking and positive fibromyalgia status were associated with greater pain and impairment among chronic pain patients, possibly as a function of depression. Although FM+ smokers report the most negative clinical symptomatology (i.e., high pain, greater interference) smoking does not appear to have a unique association with pain or functioning in FM+ patients, rather the effect is additive. The 38.7% smoking rate in FM+ patients is high, suggesting FM+ smokers present a significant clinical challenge.

Is Transcranial Direct Current Stimulation (tDCS) Effective for the Treatment of Pain in Fibromyalgia? A Systematic Review and Meta-Analysis.

PMID: 31982685 | Indexed in PubMed

Fibromyalgia is a debilitating condition characterized by chronic widespread pain. It is believed to be caused by dysfunction of the central nervous system (CNS) but current treatments are largely ineffective. Transcranial direct current stimulation (tDCS), a neuromodulation technique that targets the CNS, may offer a new line of treatment. To systematically review the most up-to-date literature and perform a meta-analysis of the effects of tDCS on pain intensity in fibromyalgia. The following databases were searched from inception: Medline (Ovid), PsychInfo, CINAHL, Cochrane Library, and Web of Science. Studies were eligible if they were randomized controlled trials, quasi-randomized trials, and nonrandomized. Crossover and parallel-group design studies were included. Risk of bias was assessed for all included studies. Meta-analysis was conducted on studies investigating pain intensity after tDCS in participants with fibromyalgia and analyzed using standardized mean difference and 95% confidence intervals. Fourteen clinical studies were included. Ten were controlled trials and 4 were within-subjects crossover studies. Meta-analysis of data from 8 controlled trials provides tentative evidence of pain reduction when active tDCS is delivered compared to sham. However, substantial statistical heterogeneity and high risk of bias of primary studies prevent more conclusive recommendations being made. tDCS is a safe intervention with the potential to lower pain intensity in fibromyalgia. However, there is a need for more empirical research of the neural target sites and optimum stimulation parameters to achieve the greatest effects before conducting further clinical studies. This systematic review and meta-analysis synthesizes current evidence for the clinical effectiveness of tDCS in the treatment of fibromyalgia pain. There is only tentative evidence of pain reduction when active tDCS is compared to sham. High heterogeneity and risk of bias across studies suggest a need for further empirical research.

State-of-the-art on fibromyalgia mechanism.

PMID: 20505625 | Indexed in PubMed

Fibromyalgia syndrome (FMS) is a very prevalent disorder defined by the presence of hyperalgesia and allodynia that leads to musculoskeletal chronic pain very frequently associated with fatigue and a diversity of other symptoms. The cause of FMS is unknown and its pathogenesis is complex and not well understood. This paper will rather discuss central pain mechanisms and the role of stressors and other factors as triggering events for onset of clinical manifestations.

[Fibromyalgia].

PMID: 11386124 | Indexed in PubMed

The classification of fibromyalgia is based on the criteria of the American College of Rheumatology. For diagnostic reasons autonomic disturbances and mental features have to be considered. The distinction between fibromyalgia (tender points) and myofascial pain syndrome (trigger points) is essential. Internal and neurological disorders as a primary cause of fibromyalgia have to be excluded. The aetiology and pathogenesis of fibromyalgia still remain uncertain. The myopathological patterns in fibromyalgia are

non-specific: type-II-fiber-atrophy, a slight increase in lipid droplets, a proliferation of mitochondria and a slightly elevated incidence of ragged red fibers. Biochemically alterations of the serotonin system and high levels of substance P in the cerebrospinal fluid of fibromyalgia patients are important. Animal experiments showed that the central stimulation by nociceptor input from muscles is exaggerated in skeletal muscle pain conditions, suggesting central hyperexcitability. The diagnosis of fibromyalgia requires a thorough exclusion of other rheumatologic and neurologic disorders. The differential diagnosis is complicated by an overlap to other chronic somatoform pain disorders.
